

```

import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.ensemble import RandomForestClassifier
from sklearn.metrics import confusion_matrix, classification_report,
accuracy_score

# Create sample dataset
app = pd.DataFrame({
    'CNT_CHILDREN': [0,1,0,2,1,0,3,2,1,0],
    'AMT_INCOME_TOTAL':
[180000,120000,250000,150000,200000,170000,300000,220000,160000,140000],
    'DAYS_BIRTH':
[-12000,-14000,-10000,-15000,-13000,-11000,-9000,-16000,-12500,-14500],
    'DAYS_EMPLOYED':
[-3000,-5000,-2500,-6000,-4000,-3500,-1500,-7000,-4500,-5500],
    'FLAG_MOBIL': [1,1,1,1,1,1,1,1,1,1],
    'FLAG_WORK_PHONE': [1,0,1,0,1,0,1,1,0,0],
    'FLAG_PHONE': [1,1,0,1,0,1,0,1,1,0],
    'FLAG_EMAIL': [0,1,0,1,1,0,1,0,1,0],
    'CNT_FAM_MEMBERS': [2,3,2,4,3,2,5,4,3,2],
    'TARGET': [1,0,1,0,1,1,0,0,1,0]
})

# Features and Target
X = app.drop(columns='TARGET')
y = app['TARGET']

# Split dataset
X_train, X_test, y_train, y_test = train_test_split(
    X,
    y,
    test_size=0.2,
    random_state=42,
    stratify=y
)

# Random Forest Function
def random_forest(X_train, X_test, y_train, y_test):

    # Initialize model
    rf_model = RandomForestClassifier(
        n_estimators=100,
        random_state=42
    )

    # Train model
    print("Training Random Forest Model...")
    rf_model.fit(X_train, y_train)

    # Predict
    print("Generating Predictions...")
    y_pred = rf_model.predict(X_test)

    # Evaluation

```

```
print("\n" + "="*40)
print("Random Forest Model Evaluation")
print("="*40)

print("\nConfusion Matrix:")
print(confusion_matrix(y_test, y_pred))

print("\nClassification Report:")
print(classification_report(y_test, y_pred, zero_division=0))

print("Accuracy:", accuracy_score(y_test, y_pred))

# Run the function
random_forest(X_train, X_test, y_train, y_test)
```